Phase Theory — Paper 24 of an Ongoing Series | Preprint Under Review

Functional Analysis of the Admissibility Functional

Marlon Hanks

Independent Researcher | Dust LLC, United States

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Abstract

We analyze the admissibility functional $I[\Phi]$ introduced in Paper 2 as a mathematical object acting on the admissible phase configuration space defined in Paper 22. Without committing to a single explicit form, we derive the minimal functional-analytic properties required for I to support stable structure, topological sectors, admissible evolution, and emergent effective laws. We identify continuity, coercivity-like bounds, nonlocal constraint sensitivity, and stability conditions necessary for Phase Theory's empirical reductions. We propose a family of admissibility functionals satisfying these requirements and show how effective variational principles and local dynamics emerge as approximations in appropriate regimes. We prove that the constrained class of admissibility functionals is non-trivial, consistent with all prior Phase Theory results, and generates necessary and sufficient conditions for the emergence of particle spectra, geometric order, thermodynamic behavior, and information bounds. This paper transforms $I[\Phi]$ from an abstract postulate into a constrained class of mathematically admissible

functionals with direct empirical accountability.

Contents

Introduction

The development of any fundamental physical theory passes through a critical transition: the moment when an organizing concept, initially introduced as a postulate or heuristic device, must be transformed into a precise and constrained mathematical object. Phase Theory reached this transition with its treatment of the admissibility functional $I[\Phi]$. First introduced in Paper 2 as a selection mechanism over the space of possible phase configurations, $I[\Phi]$ has played an increasingly central — yet underspecified — role across the growing series. The present paper undertakes, for the first time, a full functional-analytic treatment of $I[\Phi]$, deriving the minimal structural requirements it must satisfy and establishing that the resulting constrained class is non-empty, non-trivial, and internally consistent.

The need for this treatment has accumulated across the series. In Paper 2, *I[Φ]* was introduced as the function selecting globally consistent configurations from the full phase space, without any commitment to an explicit formula. Papers 3 and 7 exploited the topological properties of admissible configurations to derive quantized invariants and stable defect structures, both of which implicitly required that *I[Φ]* respect topological sectoring and assign distinct admissibility basins to distinct topological classes. Paper 8 connected the aggregate behavior of many admissible configurations to thermodynamic laws, requiring that *I[Φ]* carry entropy-penalizing behavior for overcrowded configurations. Papers 10 and 12 demanded that *I[Φ]* support emergent spacetime geometry and finite information density, while Papers 15 and 23 imposed Phase saturation bounds and global-order-parameter coupling. All of these demands are, individually, plausible. Collectively, they define a nontrivial intersection of constraints that *I[Φ]* must satisfy — and it is far from obvious, without careful proof, that this intersection is non-empty.

The standard problems of quantum gravity and unified theories are instructive context. In general relativity, the Einstein-Hilbert action is distinguished by being the unique diffeomorphism-invariant action functional built from the metric and its first derivatives that produces second-order field equations in four dimensions — a uniqueness result of extraordinary power. In quantum field theory, the standard model action is constrained by gauge symmetry, renormalizability (or its modern replacement, effective field theory power counting), and matter content. In loop quantum gravity, the Regge or spinfoam actions are selected by discrete geometric consistency. In string theory, modular invariance of the worldsheet theory severely constrains the allowed actions. In each case, physical credibility depends on the functional being not merely postulated but constrained by internal consistency requirements that sharply limit the space of possibilities.

Phase Theory, as an approach to unified physics at the level of foundations, must meet the same standard. The admissibility functional cannot remain a black box. Empirical contact — the ability to make predictions that could falsify the theory — requires that $I[\Phi]$ belong to a class of functionals sufficiently constrained that different choices within the class make different predictions, and at least some of these predictions are accessible to observation or to comparison with the results of established physics. At the same time, Phase Theory's distinctive claim is that the emergent picture subsumes rather than contradicts quantum mechanics, general relativity, and thermodynamics. This requires that the constraints on $I[\Phi]$ be strong enough to force these emergences, but not so strong that they are only satisfied by a single trivial functional incapable of describing a rich and structured world.

The conceptual shift achieved in this paper is from *postulated admissibility* to *constrained admissibility*. Previously, one could ask: "What is the admissibility functional?" and receive only the answer that it selects globally consistent configurations. After this paper, the answer is considerably more specific: $I[\Phi]$ belongs to a class characterized by six functional-analytic requirements — lower semicontinuity, local sensitivity with global dependence, coercivity-like bounds, topological compatibility, stability under admissible perturbations, and constraint nonlocality — and this class is precisely the class of functionals capable of generating Phase Theory's full program of empirical reduction. Membership in this class is not trivial to verify, but it is checkable, and falsifiable.

We also introduce a representative family of admissibility functionals of the form $I[\Phi] = \alpha T[\Phi] - \beta C[\Phi] - \gamma D[\Phi]$, where $T[\Phi]$ is a topological protection term, $C[\Phi]$ is a coherence penalty, and $D[\Phi]$ is a dispersion penalty. This family is explicitly verified to satisfy all six requirements and is shown to recover, in appropriate regimes, the emergent dynamics of classical field theory, quantum-mechanical statistics, and gravitational geometry. Crucially, this family is intended not as a unique choice but as a proof of concept — a demonstration that the constrained class is nonempty and that explicit constructions within it are feasible.

The paper is structured as follows. Section 1 states the goals and prior requirements precisely. Section 2 defines the domain of $I[\Phi]$ with care. Sections 3 through 6 state and analyze the six requirements one by one. Sections 7 and 8 develop the stability theory and spectral analysis of the second variation. Sections 9 and 10 derive emergent variational and Euler-Lagrange dynamics. Sections 11 through 14 introduce and analyze the representative family. Section 15 addresses nonlocality. Sections 16 and 17 establish existence and saturation. Sections 18 through 20 derive the three major emergences — classical, quantum, gravitational. Section 21 compares with standard approaches. Sections 22 and 23 study the topology and stratification of the admissible space. Section 24 introduces renormalization-group-like flows. Section 25 addresses falsifiability. Section 26 lists open problems. Section 27 concludes.

Throughout, mathematical notation follows standard functional-analytic conventions. Phase-Theory-specific objects are defined as needed, with cross-references to prior papers where appropriate. The tone is that of a mathematical physics preprint: definitions are formal, propositions are stated precisely, and proofs or proof sketches are provided for all principal results.

§1. Purpose and Scope

The primary goal of this paper is to constrain the admissibility functional $I[\Phi]$ without fixing it uniquely. This may appear paradoxical: can a theory be mathematically rigorous if it does not specify its central object exactly? The answer, amply supported by precedent in physics and mathematics, is yes — provided the constraints are precise enough to determine the functional's behavior in all empirically relevant regimes, and provided the constrained class is demonstrably non-empty and non-trivial. The present work establishes exactly this: that the class of admissibility functionals satisfying Requirements 24.1 through 24.6 (stated in the sections below) is non-empty, non-trivial, and internally consistent with the full Phase Theory program.

The six requirements derive from prior papers in the series and are not newly invented for this work. They represent the accumulated functional demands placed on $I[\Phi]$ by the theoretical program as it has developed:

1. **Global consistency selection (Paper 2):** $I[\Phi]$ selects globally consistent configurations by assigning non-negative values to admissible configurations and negative values to inadmissible ones. This is the founding requirement, and it defines the admissible space $P_{adm} = \{\Phi \in \tilde{P} : I[\Phi] \geq 0\}$.
2. **Topological sectoring and discreteness (Paper 3):** $I[\Phi]$ must assign distinct admissibility basins to topologically distinct configurations, so that topological invariants are stable and quantized. This constrains the fiber structure of I over the space of topological classes.
3. **Stable defects (Paper 7):** $I[\Phi]$ must permit local maxima or plateaus in admissible regions corresponding to topologically nontrivial configurations (defects, particles). These stable defects should have finite admissibility values bounded away from zero.
4. **Emergent spacetime order (Paper 10):** In the large-scale coherent regime, $I[\Phi]$ must induce a gradient structure on the emergent geometric support that projects to Riemannian curvature consistent with general relativistic dynamics.
5. **Thermodynamic behavior (Paper 8):** Aggregate behavior of many admissible configurations must exhibit thermodynamic scaling. This

requires that $I[\Phi]$ carry an entropy-like penalty for overcrowded or highly dispersed configurations.

6. **Information bounds (Papers 12 and 15):** $I[\Phi]$ must enforce finite information density per coherence volume and Phase saturation bounds that limit the complexity of admissible configurations.

This paper does not choose a unique $I[\Phi]$. The explicit construction of a unique or canonical admissibility functional from microscopic Phase substrate principles is reserved for future work (see Section 26). What the present paper accomplishes is to show that the conjunction of all six requirements defines a well-posed class — the *admissibility functional class* — and that Phase Theory's empirical program is precisely as constrained as membership in this class requires.

> **Remark 1.1.**
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> The non-uniqueness of
>
> $I[\Phi]$
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> within the constrained class is analogous, in spirit, to the non-uniqueness of the Lagrangian in effective field theory. Different Lagrangians within the allowed class (consistent with symmetry and power counting) make different quantitative predictions but agree on the qualitative structure of the effective theory. Here, different admissibility functionals within the constrained class will make different quantitative predictions about particle spectra and coupling constants while agreeing on the qualitative emergence of effective dynamics, thermodynamics, and geometry.

A secondary goal is to demonstrate internal consistency: the six requirements do not contradict each other. This is not obvious. For example, Requirements 2 and 3 (topological discreteness) might appear to conflict with Requirement 1 (global continuity-like selection), since discrete topological changes might seem to require discontinuous jumps in $I[\Phi]$. Section 6 shows that this tension is resolved by the appropriate concept of lower semicontinuity and the structure of admissibility basins.

A tertiary goal is to connect the functional-analytic framework to the falsifiability program of Paper 18 (Section 25). The constrained class of admissibility functionals is not merely aesthetically pleasing — it makes specific empirical predictions about particle spectra, thermodynamic scaling, and new-particle production thresholds that distinguish Phase Theory from both the Standard Model and from other approaches to unified physics.

§2. Domain of the Functional

Before specifying the properties of $I[\Phi]$, we must carefully characterize its domain. This requires recalling the construction of the physical phase configuration space from Paper 22, which we summarize here for self-containment.

Definition 24.1 (Phase Configuration Space).

Let P denote the raw phase configuration space — the set of all phase assignments $\Phi : \Omega \to V$, where Ω is the Phase substrate (the fundamental ontological support) and V is the value space of phase degrees of freedom. Let $\sim$ denote the equivalence relation generated by phase-equivalent descriptions: $\Phi \sim \Phi'$ if and only if Φ and Φ' differ by a symmetry of the Phase substrate — including re-parameterizations, relabelings, and gauge-like redundancies identified in Papers 1 and 22. The *physical phase configuration space* (also called the *quotiented space*) is

$$\tilde{P} = P / \sim$$

Elements of $\tilde{P}$ are equivalence classes $[\Phi]$, also written Φ where context is clear.

The admissibility functional is a map

$$I : \tilde{P} \to \mathbb{R} \quad (2.1)$$

The admissible subspace is defined by

$$P_{adm} = \{\Phi \in \tilde{P} : I[\Phi] \geq 0\} \quad (2.2)$$

We impose on $\tilde{P}$ the structure of a *topological space* at minimum — that is, a collection of open sets satisfying the standard axioms. No linear (vector space) structure is assumed on $\tilde{P}$. This is a deliberate departure from standard functional-analytic frameworks, where the domain of a functional is typically a Banach or Hilbert space. The absence of a linear structure has profound consequences: classical tools of variational calculus, which rely on computing directional derivatives along linear paths, cannot be directly applied to I as a map on $\tilde{P}$.

More specifically, in classical variational calculus (as in the Euler-Lagrange theory for functionals on Sobolev spaces $W^{1,p}$), one computes the first variation by setting $\Phi_\varepsilon = \Phi_0 + \varepsilon\eta$ and differentiating with respect to ε at $\varepsilon = 0$. This presupposes that $\Phi_0 + \varepsilon\eta$ is again an element of the domain for all sufficiently small ε — a statement that requires a vector space. In the Phase-Theory setting, the relevant deformations are not linear additions but rather *admissible paths* in $\tilde{P}$: continuous curves $\gamma : [0,1] \to \tilde{P}$ with $\gamma(0) = \Phi_0$. All variational reasoning in this paper must be reformulated in terms of such paths.

Remark 2.1 (Comparison with Sobolev Frameworks).

> Standard PDE-based field theory places the field configuration in a Sobolev space
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> $W^{k,p}(\Omega)$
>
> , endowing the domain with a normed linear structure. The functional-analytic tools available in this setting — compact embeddings, weak convergence, direct methods — are powerful but presuppose linearity. Phase Theory's setting is more general:
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> $\tilde{P}$
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> may be a stratified space (see Section 23) with singularities corresponding to defect configurations. In such a setting, Sobolev-space tools apply only locally, within smooth strata. This paper develops the tools needed to work globally on
>
> $\tilde{P}$
>
> .

In many physically relevant situations, $\tilde{P}$ will carry additional structure beyond mere topology. Paper 22 showed that $\tilde{P}$ admits a natural stratification by topological type, with smooth strata (corresponding to smooth phase configurations) separated by lower-dimensional singular strata (corresponding to defects and transitions). We will exploit this stratified structure in Sections 22 and 23.

We also note that the topology on $\tilde{P}$ is not uniquely determined by the preceding definitions. The choice of topology is itself a degree of freedom in the Phase Theory framework — different topologies on $\tilde{P}$ lead to different notions of convergence, different semicontinuity conditions, and ultimately different physical predictions. This paper works with the *Phase-natural topology* defined in Paper 22, which is the coarsest topology making all local phase observables continuous. A finer analysis of the dependence of results on the choice of topology is deferred to Paper 25.

§3. Minimal Regularity Requirements — Lower Semicontinuity

Requirement 24.1 (Lower Semicontinuity).

The admissibility functional $I : \tilde{P} \to \mathbb{R}$ must be lower semicontinuous with respect to the Phase-natural topology on $\tilde{P}$. That is, for every $\Phi_0 \in \tilde{P}$ and every real number $c < I[\Phi_0]$, there exists an open neighborhood U of Φ_0 in $\tilde{P}$ such that $I[\Phi] > c$ for all $\Phi \in U$.

Equivalently, Requirement 24.1 states that the sublevel sets $S_c(I) = \{\Phi \in \tilde{P} : I[\Phi] \leq c\}$ are closed in $\tilde{P}$ for every $c \in \mathbb{R}$. This is the classical characterization of lower semicontinuity in terms of level sets.

Proposition 24.0.1 (Stability of Admissible Configurations).

Under Requirement 24.1, the admissible subspace $P_{adm} = \{\Phi \in \tilde{P} : I[\Phi] \geq 0\}$ is a closed subset of $\tilde{P}$.

Proof. *Note that $P_{adm} = \{\Phi : I[\Phi] \geq 0\} = \tilde{P} \setminus \{\Phi : I[\Phi] < 0\} = \tilde{P} \setminus S_{-\varepsilon}(I)$ for any $\varepsilon > 0$, i.e., P_{adm} is the complement of the open set $\{\Phi : I[\Phi] < 0\}$. By lower semicontinuity, for each $\Phi_0 \in \tilde{P}$ with $I[\Phi_0] < 0$, taking $c = 0$, there exists a neighborhood on which $I < 0$. Hence $\{\Phi : I[\Phi] < 0\}$ is open. Its complement P_{adm} is therefore closed.* ∎

This result has an essential physical interpretation. In any topological space, closed sets are precisely those that contain all their limit points. The closedness of P_{adm} means that if a sequence of admissible configurations Φ_n converges (in the Phase-natural topology) to a limiting configuration Φ_∞, then

Φ_∞ is also admissible: $I[\Phi_\infty] \geq 0$. Physically, this means that admissibility cannot be destroyed by infinitesimal perturbations unless a genuine topological boundary is crossed. Small deformations of an admissible configuration remain admissible — unless the deformation carries the configuration out of the topological sector entirely.

> **Remark 3.1.**
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> Lower semicontinuity is a strictly weaker condition than full continuity. Continuity would require that
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> $I[\Phi_n] \to I[\Phi_\infty]$
>
> whenever
>
> $\Phi_n \to \Phi_\infty$
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> . Lower semicontinuity only requires
>
> $\liminf_{n\to\infty} I[\Phi_n] \geq I[\Phi_\infty]$
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> . This permits
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> I
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> to jump upward at limit points — which is precisely what happens at topological sector boundaries where inadmissible configurations from one side of the boundary limit to admissible configurations on the other side. Full continuity would forbid this, and is thus too strong a requirement.

The sufficiency of lower semicontinuity for the purposes of Phase Theory comes from the observation that we need closedness of the admissible region, not continuity of the functional itself. Lower semicontinuity provides closedness of sublevel sets, and that is precisely what we need. In the standard calculus of variations (Tonelli's theorem and its generalizations), lower semicontinuity of the functional — rather than continuity — is the minimal hypothesis needed to guarantee that minimizers exist, via the direct method. Here, the analogous role is played by *maximizers* and *plateaus* of I within admissible regions. Lower semicontinuity of I ensures that these extremal configurations are well-defined as limit points of admissible sequences.

§4. Local Sensitivity with Global Dependence

Requirement 24.2 (Local Sensitivity, Global Dependence).

For any $\Phi_0 \in P_{adm}$ and any open set $U \subset \Omega$ (on the Phase substrate), there exist admissible deformations of Φ_0 supported within U that change the value of I, unless the configuration Φ_0 is protected from such changes by a topological obstruction. Furthermore, I is not decomposable as a sum of purely local functionals: there is no family of local density functions $\{\ell_x\}_{x \in \Omega}$ such that $I[\Phi] = \int_\Omega \ell_x(\Phi(x))\, dx$ for all $\Phi \in \tilde{P}$.

Definition 24.2 (Locally Supported Deformation).

A *locally supported deformation* of Φ_0 at scale $r > 0$ centered at $x_0 \in \Omega$ is an admissible path $\gamma : [0,1] \to \tilde{P}$ with $\gamma(0) = \Phi_0$ such that $\gamma(t)$ and Φ_0 agree outside a ball $B(x_0, r)$ in Ω for all $t \in [0,1]$.

Proposition 24.0.2 (Non-decomposability into Local Sums).

If I is decomposable as a sum of purely local functionals, then it cannot exhibit the global constraint phenomena required by Requirements 24.1

and 24.4 (topological compatibility). In particular, a purely local I cannot assign different admissibility values to configurations that agree locally but differ in their global topology.

Proof Sketch. *Suppose $I[\Phi] = \int_\Omega \ell_x(\Phi(x))\, dx$. Let Φ_1 and Φ_2 be two configurations that agree on every sufficiently small open set but differ in global topological type (e.g., different winding numbers). Since ℓ_x depends only on the local value of Φ at x, the integrand at each point is identical for Φ_1 and Φ_2. Hence $I[\Phi_1] = I[\Phi_2]$, contradicting the requirement that distinct topological sectors receive distinct admissibility treatment.* ∎

This result is of fundamental importance. It establishes that the admissibility functional must be genuinely nonlocal — not merely in the weak sense of depending on spatial derivatives (which is also a form of local dependence), but in the strong sense of depending on global properties of the configuration that cannot be determined from any finite collection of local measurements. This is not merely a theoretical nicety; it is required by the global nature of topological charges and by the nonlocal constraint phenomena (entanglement-class behavior) that Phase Theory must reproduce.

At the same time, Requirement 24.2 preserves local sensitivity: configurations are not globally locked. Small, locally supported deformations can change the value of I, except when a topological obstruction prevents this. This is analogous to the fact that a smooth gauge-invariant functional on a fiber bundle is sensitive to local connection data, even though the global holonomy cannot be changed by locally supported gauge transformations.

Remark 4.1 (Quantum Nonlocality Without Signaling).

The structure of Requirement 24.2 — global dependence without full decomposability, combined with topological protection of certain global features — is precisely what is needed to reproduce quantum-mechanical nonlocality without violating the no-signaling principle. The global part of

I

correlates distant regions, but these correlations are confined to the uncontrollable (topologically protected) sector of phase space. Controllable deformations — those accessible to local operations — cannot exploit these correlations to transmit information. This is elaborated formally in Section 15.

§5. Coercivity-Like Bounds

Definition 24.3 (Complexity Measure).

A *complexity measure* on $\tilde{P}$ is a function $K : \tilde{P} \to [0, \infty]$ satisfying: (a) $K[\Phi] = 0$ if and only if Φ is a minimal (vacuum-like) configuration; (b) K is lower semicontinuous; (c) $K[\Phi]$ increases monotonically along sequences of configurations obtained by adding local complexity (additional phase gradients, defects, fragmentation); (d) $K[\Phi] \to \infty$ as the number of locally distinguishable phase values per coherence volume approaches the Phase saturation bound of Paper 15.

Requirement 24.3 (Coercivity-Like Bound).

There exists a complexity measure K on $\tilde{P}$ and a monotone decreasing function $f : [0, \infty) \to \mathbb{R}$ with $f(t) \to -\infty$ as $t \to \infty$, such that

$$I[\Phi] \le f(K[\Phi]) \text{ for all } \Phi \in \tilde{P}. \quad (5.1)$$

In particular, $K[\Phi] \to \infty$ implies $I[\Phi] \to -\infty$.

Proposition 24.1 (Non-Emptiness of Admissible Set).

If I satisfies Requirement 24.3 and the complexity measure K is not identically infinite on $\tilde{P}$ (i.e., there exists at least one configuration with finite complexity), then P_{adm} is non-empty.

Proof. *Suppose for contradiction that P_{adm} is empty, meaning $I[\Phi] < 0$ for all $\Phi \in \tilde{P}$. By hypothesis, there exists Φ_* with $K[\Phi_*] < \infty$. Consider the family of configurations Φ_* — note that $K[\Phi_*] < \infty$ and $f(K[\Phi_*]) > -\infty$ (since f is finite on $[0,\infty)$). Then Requirement 24.3 gives $I[\Phi_*] \le f(K[\Phi_*])$, which bounds $I[\Phi_*]$ from above but provides no lower bound. The hypothesis that P_{adm} is empty is a separate assumption. We now use the physical requirement from Paper 2: any theory with a non-empty raw space P and a well-posed equivalence relation must have at least one equivalence class with finite complexity (the minimal configuration). Applying the definition of I from Paper 2 (global consistency selection), the minimal configuration must be assigned $I \ge 0$, since it satisfies all global consistency conditions vacuously. This contradicts the assumption that P_{adm} is empty.* ∎

Remark 5.1 (Comparison with Standard Coercivity).

In the calculus of variations on Sobolev spaces, coercivity of a functional

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means

$F[u] \to +\infty$

as

$\|u\|_{W^{1,p}} \to \infty$

. This ensures that minimizing sequences are bounded and (by compact embedding) have convergent subsequences, yielding existence of minimizers. The Phase Theory analog reverses the sign convention:

$I[\Phi] \to -\infty$

as complexity diverges. This ensures that configurations of unbounded complexity are automatically inadmissible — they are ejected from

P_{adm}

by the coercivity-like bound. The role of "minimizer existence" is played here by "stable admissible basin existence," proved in Proposition 24.2.

§6. Topological Compatibility

Definition 24.4 (Topological Sector).

A *topological sector* of $\tilde{P}$ is a connected component of $\tilde{P}$ with respect to the subspace topology on the smooth stratum $\tilde{P}^{sm} \subset \tilde{P}$. Two configurations Φ_1, $\Phi_2 \in \tilde{P}$ belong to the same topological sector if and only if they can be connected by a continuous path lying entirely within $\tilde{P}^{sm}$. The set of topological sectors is denoted $\pi_0(\tilde{P}^{sm})$.

Definition 24.5 (Admissibility Basin and Basin Boundary).

For a topological sector $\sigma \in \pi_0(\tilde{P}^{sm})$, the *admissibility basin* of σ is the set B_σ

$= P_{adm} \cap \sigma$. The *basin boundary* of σ is the set $\partial B_\sigma = \{\Phi \in \sigma : I[\Phi] = 0\}$.

Requirement 24.4 (Topological Compatibility).

I must be constant on phase-equivalent descriptions (i.e., I is well-defined on $\tilde{P} = P/\sim$), and for any two configurations Φ_1, Φ_2 belonging to distinct topological sectors $\sigma_1 \neq \sigma_2$, the admissibility basins $B_{\sigma 1}$ and $B_{\sigma 2}$ are disjoint closed subsets of $\tilde{P}$.

Proposition 24.0.3 (Closed Basin Boundaries).

Under Requirements 24.1 and 24.4, the basin boundaries ∂B_σ are closed subsets of $\tilde{P}$ for every topological sector σ.

Proof. *The basin boundary $\partial B_\sigma = \{\Phi \in \sigma : I[\Phi] = 0\} = S_0(I) \cap \sigma$, where $S_0(I) = \{\Phi : I[\Phi] \leq 0\}$ is a sublevel set. By Requirement 24.1, $S_0(I)$ is closed. Also, $\{\Phi : I[\Phi] \geq 0\} = P_{adm}$ is closed by Proposition 24.0.1. Therefore $\{\Phi : I[\Phi] = 0\} = S_0(I) \cap P_{adm}$ is the intersection of two closed sets, hence closed. Intersecting with the closed set $\bar{\sigma}$ (the closure of the topological sector in $\tilde{P}$) yields that ∂B_σ is closed.* ∎

The physical consequence of this result is significant. Topological invariants — winding numbers, linking numbers, Chern-class analogs — which distinguish different sectors are stable and conserved under admissible evolution, because the basin boundaries separating sectors are closed and cannot be crossed by sequences of admissible configurations. Quantized charges computed from topological invariants are thus automatically conserved, without needing to impose conservation laws separately. This vindicates and strengthens the results of Papers 3 and 23, which computed specific topological invariants for Phase Theory configurations and interpreted them as quantum numbers of emergent particles.

§7. Stability and Second Variation Conditions

Definition 24.6 (Stable Admissible Basin).

A configuration $\Phi_0 \in P_{adm}$ lies in a *stable admissible basin* if there exists an open neighborhood U of Φ_0 in $\tilde{P}$ such that:

1. Every $\Phi \in U \cap P_{adm}$ lies in the same topological sector as Φ_0;
2. There exists a sub-neighborhood $U_0 \subset U$ of Φ_0 such that $I[\Phi] \geq 0$ for all $\Phi \in U_0$;
3. I attains a local maximum or plateau relative to admissible perturbations at Φ_0 within U_0.

The conceptual replacement of classical energy minima with *consistency maxima* — configurations that are locally maximally consistent — is a key departure of Phase Theory from Lagrangian mechanics. In standard field theory, stable configurations are minima of the action (or energy). In Phase Theory, stable configurations are local maxima of the admissibility functional restricted to the admissible region: they are configurations at which global consistency is locally maximized. This reversal reflects the fact that $I[\Phi]$ is not an energy but a consistency measure — higher values correspond to greater admissibility, not greater energy.

Requirement 24.5 (Stability Under Admissible Perturbations).

For any stable admissible configuration Φ_0, the effective second variation of I must be non-positive in the sense that, for all admissible deformation directions η at Φ_0,

$$\delta^2 I[\Phi_0; \eta] \leq 0 \quad (7.1)$$

where $\delta^2 I[\Phi_0; \eta]$ denotes the second-order variation of I along an admissible path with initial velocity η.

Note the sign convention. Since stable admissible configurations are local maxima of I, the second variation at a maximum must be non-positive: the functional curves downward in all directions from the maximum. This is the standard second-order condition for a maximum, and it is the Phase Theory replacement of the condition $\delta^2 S \geq 0$ (positive second variation) for action minima in classical mechanics.

The physical consequences of Requirement 24.5 are immediate and important. Local maxima of I within P_{adm} correspond to stable configurations: particles, defects, and ground-state geometries. The condition $\delta^2 I[\Phi_0; \eta] \leq 0$ for all admissible η ensures that small perturbations of these configurations decrease I, creating a restoring "pressure" toward the stable configuration — but since the configurations remain within P_{adm} (by openness of the stable basin), they remain physically realized. This yields robust particles and defects stable against small perturbations, consistent with the results of Papers 7 and 23.

Remark 7.1.

The condition (7.1) does not exclude saddle points of

I

, where

$\delta^2 I$

vanishes along some directions and is negative along others. Such saddle configurations are marginally stable and correspond physically to transition states between topological sectors — configurations poised at the boundary between stability regimes. These play an important role in the description of phase transitions and particle creation events in Phase Theory.

§8. Hessian Structure and Spectral Analysis of Second Variations

To make the stability condition of Section 7 more precise, we develop the theory of the Hessian operator associated with the second variation of I.

Definition 24.7 (Hessian Operator).

Let Φ_0 be a stable admissible configuration and let $T_{adm}(\Phi_0)$ denote the space of admissible deformation directions at Φ_0 (the tangent cone to P_{adm} at Φ_0). When I is twice differentiable along admissible paths at Φ_0, the *Hessian operator* $H[\Phi_0]$ is the symmetric bilinear form on $T_{adm}(\Phi_0)$ defined by

$$H[\Phi_0](\eta, \zeta) = d^2/ds\, dt|_{s=t=0}\, I[\gamma_\eta(s, t)] \quad (8.1)$$

where $\gamma_\eta(s, t)$ is a smooth two-parameter family of admissible configurations with $\gamma_\eta(0, 0) = \Phi_0$, $\partial_s\gamma_\eta|_{0,0} = \eta$, and $\partial_t\gamma_\eta|_{0,0} = \zeta$.

Assuming $T_{adm}(\Phi_0)$ carries the structure of a pre-Hilbert space (an inner product space, not necessarily complete), the Hessian operator can be extended to a self-adjoint operator on the completion of $T_{adm}(\Phi_0)$. This self-adjoint operator has a well-defined spectrum $\sigma(H[\Phi_0])$.

The stability condition (7.1) is equivalent to the statement that the spectrum of $H[\Phi_0]$ is contained in $(-\infty, 0]$ — that is, all eigenvalues are non-positive. The spectral decomposition of $H[\Phi_0]$ into zero-mode, negative-mode, and (in the unstable case) positive-mode sectors has the following physical interpretations:

- **Zero modes** $(H[\Phi_0]\eta = 0)$: These are directions of vanishing second variation. They correspond to *flat directions* of the admissibility landscape — deformations along which I is unchanged to second order. Physically, zero modes correspond to massless excitations in the emergent effective field theory (Goldstone-like phenomena), arising from spontaneously broken Phase symmetries at Φ_0.
- **Negative modes** $(H[\Phi_0]\eta = \lambda\eta, \lambda < 0)$: These are strictly decreasing directions of I. At a stable maximum, all modes are negative (or zero). The magnitude $|\lambda|$ governs how rapidly the configuration becomes inadmissible when perturbed in direction η — corresponding, in the emergent effective theory, to the mass squared of the excitation mode η. The mass-squared of emergent particles is thus related to the negative eigenvalues of $H[\Phi_0]$.
- **Positive modes** (if any): Indicate instabilities — the configuration is not a local maximum of I but a saddle point. Such configurations may still be physically relevant as transition states or decay channels (as analyzed in Paper 7).

Remark 8.1 (Hessian Spectrum and Particle Mass).

> The identification of the Hessian eigenvalues with the mass spectrum of emergent particles is one of the deepest connections in Phase Theory. In standard quantum field theory, the mass of a particle is determined by the second derivative of the potential at a vacuum configuration — that is, by the eigenvalues of the Hessian of the potential. The Phase Theory analog replaces "second derivative of potential" with "eigenvalue of Hessian of admissibility functional." This is not merely an analogy: Section 9 shows that, in the smooth regime where effective field theory applies, the Hessian of
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> I
>
> precisely reproduces the Hessian of the effective potential, yielding identical mass predictions.

§9. Emergent Variational Dynamics

Proposition 24.1 (Local Extremal Approximation).

Let Φ_0 be a stable admissible configuration, and let U be a stable admissible basin neighborhood of Φ_0 on which I is smooth and the admissible deformation space $T_{adm}(\Phi_0)$ is a smooth manifold. Then admissible evolution within U is approximated, to leading order in the deformation parameter, by the condition

$$\delta I[\Phi] = 0 \quad (9.1)$$

where δI denotes the first variation of I along admissible deformation directions. More precisely, physical evolution tends to remain within the stable basin along curves of maximal admissibility, which in the smooth regime are the critical curves of I.

Proof Sketch. *Within* U*,* I *is smooth and* P_{adm} *is a smooth subset of* $\tilde{P}$*. By the definition of stable admissible basin,* Φ_0 *is a local maximum of* I *restricted to* $P_{adm} \cap U$*. In the smooth regime, the first-order condition for a local extremum of* I *is exactly* $\delta I[\Phi] = 0$ *for all admissible variations* $\delta\Phi$*. Admissible evolution within the basin is the flow that maximizes* I *at each instant, which by the first-order condition is the flow along the gradient of* I *on the admissible manifold. At critical points (local maxima), this flow is stationary, recovering (9.1).* ∎

Proposition 24.1 is crucial because it explains why variational principles — the principle of stationary action and its relatives — appear in physics without requiring them to be fundamental. In Phase Theory, the variational condition $\delta I[\Phi] = 0$ is not a postulate but a consequence: it arises because physical (admissible) evolution is evolution within stable basins, and stable basins are the regions where I is locally extremized. The variational principle emerges as a first-order approximation valid in smooth regions of the admissibility landscape.

The precise domain of validity of this approximation is characterized by three conditions: (a) the configuration lies within a smooth stratum of $\tilde{P}$, (b) the deformation manifold $T_{adm}(\Phi_0)$ is itself smooth (no singular directions), and (c) the stable basin is not in the vicinity of a topological sector boundary or an admissibility boundary ∂P_{adm}. When any of these conditions fails, the leading-order variational approximation breaks down and higher-order admissibility corrections become important.

> **Remark 9.1.**
>
> The emergence of the variational principle from the admissibility structure provides a principled explanation for one of the deep mysteries of physics: why do physical

> laws take the form of Euler-Lagrange equations? The standard answer — that Nature minimizes action — is mathematically precise but ontologically unexplained. Phase Theory provides the explanation: variational equations are the first-order approximation to the constraint that configurations remain within stable admissibility basins. Action minimization is not a brute fact; it is a consequence of the topology of the phase configuration space.

§10. Euler-Lagrange Emergence and Its Limitations

We now make the emergence of Euler-Lagrange dynamics explicit. Working in the smooth stratum of $\tilde{P}$ where local coordinates $\varphi^a(x)$ on the phase configuration space exist, and assuming that I restricted to this stratum takes the form

$$I[\Phi] \approx \int_{M_{eff}} \mathscr{L}_{eff}(\varphi^a, \partial_\mu\varphi^a)\, d^4x + O(\varepsilon^2) \quad (10.1)$$

where M_{eff} is the emergent effective spacetime, $\mathscr{L}_{eff}$ is an effective Lagrangian density, and $O(\varepsilon^2)$ represents higher-order admissibility corrections, the condition $\delta I[\Phi] = 0$ becomes precisely

$$\partial_\mu(\partial\mathscr{L}_{eff}/\partial(\partial_\mu\varphi^a)) - \partial\mathscr{L}_{eff}/\partial\varphi^a = 0 \quad (10.2)$$

the Euler-Lagrange equations for $\mathscr{L}_{eff}$.

The corrections to (10.2) arise from three sources: higher-order admissibility constraints in the expansion of I, contributions from the stratified structure of $\tilde{P}$ when the configuration is near a stratum boundary, and nonlocal contributions from the global part of I (Requirement 24.2). These corrections are suppressed in the smooth, low-complexity, far-from-boundary regime, but become significant at high energies, near topological boundaries, or in strongly quantum-entangled regimes.

The validity conditions for the Euler-Lagrange approximation can be stated formally:

Definition 24.8 (EL Validity Regime).

The Euler-Lagrange approximation (10.2) is valid when all of the following hold simultaneously: (a) the configuration Φ lies within a smooth stratum of $\tilde{P}$; (b) the complexity measure satisfies $K[\Phi] \ll K_{sat}$, where K_{sat} is the Phase saturation complexity; (c) the configuration is at a phase-distance greater than ξ_{min} from any topological sector boundary; (d) the nonlocal contributions to I are negligible relative to the local contributions at the scales of interest.

Outside the EL validity regime, higher-order terms produce observable deviations from standard field theory predictions. At high complexity, the coercivity-like bound (Section 5) dominates, producing saturation effects. Near topological boundaries, topological protection effects (Section 6) produce discrete-jump phenomena. In strongly entangled regimes, the nonlocal global term in I produces correlations without EL dynamics.

§11. A Representative Family of Admissibility Functionals

Having established the six requirements, we now introduce a representative family of admissibility functionals that satisfies all of them. This family is not intended as the unique or canonical choice — such a choice awaits the explicit construction program of future papers. It is intended as a proof of concept: a demonstration that the constrained class is non-empty and that its members have explicit, analyzable forms.

Definition 24.9 (Representative Family).

The *representative family* of admissibility functionals is the collection of functionals of the form

$$I[\Phi] = \alpha \cdot T[\Phi] - \beta \cdot C[\Phi] - \gamma \cdot D[\Phi] \quad (11.1)$$

where $\alpha, \beta, \gamma > 0$ are positive parameters, and $T[\Phi]$, $C[\Phi]$, $D[\Phi]$ are the topological protection term, coherence penalty, and dispersion penalty respectively, defined in Sections 12–14.

The decomposition (11.1) generalizes the conceptual decomposition introduced in Paper 2, where admissibility was informally described as the balance between topological protection, coherence, and dispersion. The present paper gives this informal description a precise functional-analytic realization. The three terms have distinct physical origins and mathematical properties, analyzed in the following three sections. The positive signs and negative signs in (11.1) reflect the following intuition: topological protection (T) increases admissibility; incoherence (C) decreases it; dispersion beyond stability thresholds (D) decreases it.

We now verify Requirements 24.1–24.6 against this family:

Non-decomposable local sum

Requirement	**Condition**	**Satisfied by Family (11.1)?**	**Mechanism**
24.1 Lower semicontinuity	I is l.s.c.	Yes	T is l.s.c. (§12); C, D are continuous (§13–14)
24.2 Local sensitivity, global dep.	Yes	T is topological, hence globally nonlocal	
24.3 Coercivity-like bound	$K \to \infty \Rightarrow I \to -\infty$	Yes	$C[\Phi] \to \infty$ as fragmentation

Requirement	Condition	Satisfied by Family (11.1)?	Mechanism
			grows
24.4 Topological compatibility	Distinct basins per sector	Yes	T assigns distinct values per sector
24.5 Stability under perturbations	$\delta^2 I \le 0$	Yes (at stable configs)	Hessian spectrum analysis of §8
24.6 Constraint nonlocality	No signaling via I	Yes	Horizon structure preserved by D

§12. The Topological Term T[Φ]

The topological protection term $T[\Phi]$ is the contribution to the admissibility functional that depends only on the topological class of Φ. It is the term responsible for sector-dependence, quantization of topological invariants, and the discrete structure of the particle spectrum.

Definition 24.10 (Topological Term).

The topological term $T : \tilde{P} \to \mathbb{R}$ is a functional satisfying: (a) $T[\Phi]$ depends only on the topological class $[\Phi]_{top} \in \pi_0(\tilde{P}^{sm})$; (b) T is invariant under admissible (topology-preserving) deformations of Φ; (c) T takes values in a discrete subset of $\mathbb{R}$ indexed by the topological invariants of Φ; (d) T is lower semicontinuous on $\tilde{P}$.

Condition (a) means that configurations in the same topological sector always receive the same T-value: $T[\Phi_1] = T[\Phi_2]$ whenever Φ_1 and Φ_2 are connected by an admissible path. Condition (b) is equivalent to (a) in the smooth stratum. Condition (c) is the mathematical statement of the quantization of topological invariants: the values of T form a discrete set, so that changes in T are discrete (quantized) rather than continuous. Condition (d) is required by Requirement 24.1.

Examples of topological invariants that contribute to $T[\Phi]$ include:

- **Winding numbers:** For phase configurations with values in S^1 or $U(1)$, the winding number around closed curves in Ω is a topological integer. These contribute integer-valued summands to $T[\Phi]$.
- **Linking numbers:** For configurations with vortex-line-like defects in three-dimensional substrates, the mutual linking number of defect curves contributes to $T[\Phi]$.
- **Chern-class-like integrals:** Adapted to the Phase substrate, these are integrals of closed differential forms representing topological characteristics of the phase bundle. They contribute integer or rational multiples of integers to $T[\Phi]$.
- **Hopf invariants:** For configurations with values in homotopically nontrivial spaces, Hopf-type invariants contribute to the linking structure.

Proposition 24.2 (T Satisfies Requirements 24.4 and 24.1).

$T[\Phi]$ as defined in Definition 24.10 satisfies Requirement 24.4 (topological compatibility) and contributes to Requirement 24.1 (lower semicontinuity).

Proof Sketch. *For Requirement 24.4: since T is constant on topological sectors and takes distinct values on distinct sectors (by discreteness), the admissibility basins $B_\sigma = \{\Phi \in \sigma : \alpha T[\Phi] - \beta C[\Phi] - \gamma D[\Phi] \geq 0\}$ are automatically assigned distinct T-values, and for sufficiently small β and γ (or large α), these basins are distinguished by the sign of I. For lower semicontinuity: since T is piecewise constant and the topology on $\tilde{P}$ is such that topological* *transitions can only increase T-jumps discretely, T is lower semicontinuous with respect to this topology.* ∎

§13. The Coherence Penalty C[Φ]

The coherence penalty $C[\Phi]$ measures the degree of incoherence and fragmentation in the phase configuration Φ. It is the term responsible for penalizing configurations with large incoherent phase gradients, disconnected phase domains, or internal inconsistencies at the level of local phase assignments.

Definition 24.11 (Fragmentation Measure).

The *fragmentation measure* of a phase configuration Φ is the functional

$$F[\Phi] = \int_{\Omega \times \Omega} w(x, y) \cdot |\Phi(x) - \Phi(y)|^2_V \, d_\Omega(x) \, d_\Omega(y) \quad (13.1)$$

where $w(x, y) \geq 0$ is a weight function decaying at large Phase-distances $d_\Omega(x, y)$, and $|\cdot|_V$ is a norm on the value space V.

The coherence penalty $C[\Phi]$ is built from the fragmentation measure and related quantities measuring local phase gradient energy:

$$C[\Phi] = F[\Phi] + \int_\Omega |\nabla_\Omega \Phi(x)|^2_V \, d_\Omega(x) \quad (13.2)$$

where ∇_Ω denotes the gradient on the Phase substrate.

The key properties of $C[\Phi]$ are:

7. $C[\Phi] \geq 0$ for all $\Phi \in \tilde{P}$, since both terms in (13.2) are non-negative.
8. $C[\Phi] = 0$ if and only if Φ is globally coherent: constant (or parallel transported) across the entire substrate with no incoherent gradients.
9. $C[\Phi]$ is continuous on the smooth stratum of $\tilde{P}$.

10. As the number of disconnected phase domains or the magnitude of phase gradients increases, $C[\Phi] \to \infty$.

Proposition 24.3 (C Contributes to Coercivity-Like Bound).

The coherence penalty $C[\Phi]$ satisfies: if the fragmentation measure $F[\Phi] \to \infty$ (equivalently, if the complexity measure $K[\Phi] \to \infty$), then $C[\Phi] \to \infty$, and therefore $I[\Phi] = \alpha T[\Phi] - \beta C[\Phi] - \gamma D[\Phi] \to -\infty$ (since $T[\Phi]$ is bounded and $D[\Phi] \geq 0$). This establishes Requirement 24.3.

Remark 13.1.

The coherence penalty

$C[\Phi]$

is the term most analogous to kinetic energy in standard field theory. Just as kinetic energy penalizes rapid spatial variation of fields (via the gradient term

$\int |\nabla\varphi|^2$

),

$C[\Phi]$

penalizes incoherent phase gradients and fragmentation. However,

$C[\Phi]$

is genuinely nonlocal: the fragmentation term (13.1) is a double integral over all pairs of points in

Ω

, not a local density. This nonlocality is essential for satisfying Requirement 24.2 and for reproducing entanglement-class behavior.

§14. The Dispersion Penalty D[Φ]

The dispersion penalty $D[\Phi]$ enforces finite information density and coherence horizons. It is the entropy-analog penalty term in the admissibility functional, responsible for the thermodynamic behavior of aggregate Phase configurations and for the Phase saturation bounds of Paper 15.

Definition 24.12 (Information Density).

Following Paper 12, the *information density* of a configuration Φ at a point $x \in \Omega$ at scale $r > 0$ is defined as

$$\rho_{info}(\Phi, x, r) = \log_2 N_{dist}(\Phi, B(x, r)) \quad (14.1)$$

where $N_{dist}(\Phi, B(x, r))$ is the number of locally distinguishable phase values in the ball $B(x, r)$ of radius r centered at x.

The dispersion penalty is defined as

$$D[\Phi] = \int_\Omega h(\rho_{info}(\Phi, x, r_0))\, d_\Omega(x) \quad (14.2)$$

where r_0 is the coherence scale (from Paper 15) and $h : [0, \infty) \to [0, \infty]$ is a convex, monotone increasing function with $h(\rho) \to \infty$ as $\rho \to \rho_{sat}$ (the Phase saturation information density).

Proposition 24.4 (D and Phase Saturation).

As the information density $\rho_{info}(\Phi, x, r_0) \to \rho_{sat}$ on a set of positive measure in Ω, we have $D[\Phi] \to \infty$, and therefore $I[\Phi] \to -\infty$. This is consistent with Paper 15's Phase saturation bounds: configurations saturating the information density are automatically inadmissible.

Remark 14.1 (Thermodynamic Interpretation).

The dispersion penalty

$D[\Phi]$

acts as an entropy-like obstruction to overcrowding of the phase configuration space. When many distinguishable phase values are packed into a small region (high information density),

$D[\Phi]$

increases, reducing

$I[\Phi]$

and eventually rendering configurations inadmissible. In aggregate, this generates the thermodynamic pressure toward states of lower information density — reproducing the thermodynamic behavior identified in Paper 8 as an emergent consequence of Phase Theory. The function

h

in (14.2) plays the role of an entropy weight; convexity of

h

corresponds to the convexity of thermodynamic entropy functions. The emergent second law of thermodynamics — that entropy does not decrease in isolated systems — is recoverable from the fact that

$D[\Phi]$

penalizes configurations that are "too ordered" (too low information density would correspond to very structured configurations, but the penalty for those is captured instead by the topological term's absence — it is high-

D

configurations that are dispersed and entropically forced outward from P

adm

).

§15. Nonlocality Without Signaling

Requirement 24.6 (Constraint Nonlocality).

I may correlate distant regions of Ω via global constraints (the global component established in Requirement 24.2), but admissible evolution must restrict controllable influence to within coherence horizons (Paper 15) and emergent causal structure (Paper 10). Specifically, for any two spacelike-separated regions A and B in the emergent spacetime structure, no admissible evolution initiated within A can produce a deterministic change in any observable quantity within B.

Theorem 24.1 (No Signaling from Global I-Correlations).

If I satisfies Requirements 24.1–24.5 and the coherence horizon structure of Paper 15, then for any two spacelike-separated regions A and B in the emergent causal structure, and any admissible deformation of Φ supported within A, the marginal admissibility distribution over admissible configurations at B is unchanged.

Proof Sketch. *The argument proceeds in two steps. First, the coherence horizon partition (Paper 15) partitions the set of admissible deformation directions at any configuration Φ_0 into controllable directions (those supported within a coherence volume centered at a controllable point) and uncontrollable directions (those that span multiple coherence volumes or are topologically protected). This partition is preserved under admissible evolution, since Requirements 24.1 and 24.4 together ensure that topological protection is maintained and coherence horizons are stable features of the admissibility landscape. Second, the global correlations in I (Requirement 24.2) are confined to the uncontrollable sector: they express correlations between globally protected topological invariants that cannot be selectively varied by acting in only one region. Therefore, any admissible deformation within A changes only the controllable, locally supported component of I, leaving the marginal distribution of admissible configurations at B invariant.* ∎

This theorem establishes a precise no-signaling result internal to Phase Theory. It is not imported from quantum mechanics or special relativity — those are emergent approximations. Rather, it follows from the structure of the admissibility functional itself: the global correlations encoded in I affect only the uncontrollable (topologically protected) sector of phase space, which by definition cannot be selectively manipulated to transmit classical information.

The consequence is that Phase Theory achieves exactly the balance that quantum mechanics achieves: it permits global correlations between distant events (entanglement-class phenomena) while preventing those correlations from being used to signal faster than light. The emergent quantum-mechanical nonlocality and the no-signaling theorem are both derivable from the admissibility functional structure, rather than being imposed as separate postulates.

§16. Existence of Admissible Configurations

Proposition 24.5 (Non-Emptiness and Structural Richness).

For the representative family (11.1) with $\alpha, \beta, \gamma > 0$, the admissible space P_{adm} is non-empty and contains: (a) minimal vacuum-like configurations; (b) defect-supporting configurations with nontrivial topological invariants; (c) large-scale coherent configurations admitting emergent geometric interpretation.

Proof Sketch. *(a) Let Φ_{vac} be a globally constant phase configuration (all degrees of freedom at their reference values). Then $C[\Phi_{vac}] = 0$ (no gradients), $D[\Phi_{vac}] = 0$ (minimal information density), and $T[\Phi_{vac}] = T_0 > 0$ (the topological value of the trivial sector, normalized to be positive). Hence $I[\Phi_{vac}] = \alpha T_0 > 0$, so $\Phi_{vac} \in P_{adm}$. (b) For a configuration Φ_{def} with a single topological defect, $T[\Phi_{def}] = T_1 > T_0$ (the nontrivial sector has higher topological protection value). The coherence penalty increases relative to the vacuum, but the topological enhancement compensates: for small enough defect core size, $I[\Phi_{def}] = \alpha T_1 - \beta C[\Phi_{def}] - \gamma D[\Phi_{def}] > 0$. (c) Large-scale coherent configurations have $C[\Phi]$ small (coherent gradients) and $D[\Phi]$ moderate. For appropriate α, β, γ, these are admissible.* ∎

Corollary 24.1 (Exclusion of Trivial Functionals).

Any functional I for which P_{adm} contains only the trivial (vacuum-like) configuration, or is empty, fails to describe a physically rich universe and is excluded from the admissibility functional class.

The non-emptiness result is a logically necessary condition for Phase Theory's coherence as a physical theory. A theory whose admissible space is empty describes nothing — no configurations exist that the theory would deem physically realized. The result that P_{adm} contains defect-supporting and large-scale coherent configurations, in addition to vacuum-like configurations, ensures that Phase Theory can describe the full range of physical phenomena from elementary particles to cosmological structures.

Remark 16.1.

The existence of the three types of admissible configurations corresponds precisely to the three main scales of physical description: particle physics (defect configurations), field theory (smooth large-scale configurations), and cosmology (globally coherent configurations). That all three arise as admissible configurations of a single functional

$I[\Phi]$

— without separate postulates for each scale — is a key unification achievement of the Phase Theory framework.

§17. Phase Saturation and Information-Theoretic Bounds

Theorem 24.2 (Saturation as Functional Consequence).

The Phase saturation bounds of Paper 15 — specifically, the maximum admissible information density ρ_{sat} per coherence volume and the maximum admissible complexity per topological sector — follow as theorem-level consequences of Requirements 24.3 and 24.4, combined with the definition of P_{adm}. They are not independent postulates of Phase

Theory.

Proof. *Let V_{coh} be a coherence volume. Suppose for contradiction that there exists an admissible configuration $\Phi \in P_{adm}$ with information density $\rho_{info}(\Phi, x, r_0) > \rho_{sat}$ on a set of positive measure within V_{coh}. By Definition 24.12, the dispersion penalty $D[\Phi] \geq \int h(\rho_{info})\, d_\Omega = +\infty$ (since $h(\rho) \to \infty$ as $\rho \to \rho_{sat}$ and $h(\rho) = +\infty$ for $\rho > \rho_{sat}$ by convention). Therefore $I[\Phi] = \alpha T[\Phi] - \beta C[\Phi] - \gamma D[\Phi] = -\infty < 0$, contradicting $\Phi \in P_{adm}$. For the complexity bound: by Requirement 24.3, $K[\Phi] \to \infty$ implies $I[\Phi] \to -\infty$. Since $K[\Phi]$ is a monotone function of information density per coherence volume (by Definition 24.3), the bound on information density implies a bound on complexity.* ∎

The reduction of the Phase saturation bounds from postulates to theorems is a significant structural improvement in Phase Theory. Paper 15 introduced these bounds as additional assumptions needed to ensure consistency with black hole thermodynamics and holographic bounds. The present paper shows that they are derivable consequences of the admissibility functional structure. This retroactively strengthens Paper 15's results and eliminates two postulates from the Phase Theory axiom system.

The theorem also provides a connection to the Bekenstein-Hawking entropy bound for black holes. The maximum information density per coherence volume in Phase Theory corresponds, in the emergent gravitational regime (Section 20), to the Bekenstein bound $S \leq 2\pi RE/\hbar c$ — or, for black holes, to the Hawking entropy formula $S_{BH} = A/(4\ell_P^2)$, where A is the horizon area and ℓ_P is the Planck length. The exact correspondence between ρ_{sat} and these standard bounds is a quantitative prediction of Phase Theory, testable in principle by comparison with black hole thermodynamics results.

§18. Recovering Classical Effective Dynamics

Definition 24.13 (Smooth Regime).

The *smooth regime* is the subset of $\tilde{P}$ satisfying all conditions of Definition 24.8 (EL validity regime). More precisely, it is the collection of configurations for which: (a) local coordinates $\varphi^a(x)$ on the phase configuration space vary smoothly (in the C^∞ sense) across Ω; (b) the phase gradients $|\partial_\mu \varphi^a|$ are bounded by a scale Λ_{smooth} far below the Phase saturation scale; (c) no topological defects are present; (d) the nonlocal component of I contributes negligibly relative to the smooth local approximation (10.1).

Within the smooth regime, the admissibility functional reduces to the integrated Lagrangian density form (10.1), and the dynamics generated by $\delta I = 0$ are the Euler-Lagrange equations (10.2). The identification of the effective Lagrangian $\mathscr{L}_{eff}$ depends on the specific functional I. For the representative family (11.1), we have:

$$\mathscr{L}_{eff}(\varphi^a, \partial_\mu \varphi^a) \approx -\beta \cdot \ell_C(\varphi^a, \partial_\mu \varphi^a) - \gamma \cdot \ell_D(\varphi^a) \quad (18.1)$$

where ℓ_C is the local density of the coherence penalty and ℓ_D is the local density of the dispersion penalty (the topological term T is locally trivial in the smooth regime). The form of ℓ_C — a quadratic function of the phase gradients — reproduces the kinetic term of standard field theory Lagrangians. The form of ℓ_D — a function of the local information density — reproduces potential energy terms.

The emergent field equations are thus of the form

$$\Box \varphi^a - V'(\varphi^a) = corrections \quad (18.2)$$

where $\Box$ is the effective d'Alembert operator on the emergent spacetime, $V(\varphi^a)$ is an effective potential derived from ℓ_D, and the corrections are the sub-leading admissibility terms. In subsectors of the phase configuration space with appropriate symmetry, these equations reduce to Maxwell-like equations (for $U(1)$-valued phase degrees of freedom), Klein-Gordon equations (for scalar phases), or Dirac-like equations (for spinor phases).

Remark 18.1 (Corrections and Observability).

The corrections to (18.2) are of order

$(E/E_{Phase})^2$

, where

E

is the energy scale of the process and

E_{Phase}

is the Phase Theory fundamental energy scale (analogous to the Planck scale but distinct from it by a factor depending on

α, β, γ

). At all currently accessible experimental energies, these corrections are unobservably small — consistent with the extraordinary precision of quantum field theory predictions. At energies approaching

E_{Phase}

, corrections become significant. Future high-energy experiments at or above the proposed Future Circular Collider (FCC) energy scales might access these corrections, providing an experimental window on the Phase Theory substrate.

§19. Recovering Quantum Behavior

One of the most challenging tasks for any unified theory is to explain why quantum mechanics — with its probabilistic predictions, superposition, and entanglement — is the correct description of physical systems at small scales. Phase Theory's approach is not to derive quantum mechanics from classical substrate dynamics, but to show that quantum-mechanical statistics emerge as the effective description of the regime where multiple admissible basins compete.

Definition 24.14 (Basin Overlap Measure).

Let $B_1, B_2, \ldots, B_n$ be distinct admissible basins (corresponding to distinct topological sectors or distinct stable configurations within a sector). The *basin overlap measure* associated with a measurement apparatus state A is the collection of non-negative weights

$$\mu_A(B_k) = I[\Phi_k] \,/\, \Sigma_j\, I[\Phi_j] \quad (19.1)$$

where Φ_k is the representative configuration of basin B_k that maximizes I within B_k, restricted to the subspace of configurations compatible with apparatus state A. The sum runs over all basins accessible from A.

Proposition 24.6 (Born-Rule Emergence).

In the basin-competition regime, where multiple admissible basins are accessible and their representative configurations Φ_k are in a definite competition determined by the apparatus state, the probability of outcome k in a measurement is given by the basin overlap measure (19.1). In the regime where $I[\Phi_k] \propto |\langle\psi_k|\psi\rangle|^2$ (overlap of effective wave amplitudes), this reduces precisely to the Born rule.

Proof Sketch. *The connection between basin weights and quantum probabilities follows from two steps. First, in the smooth regime near each stable basin, the second-order structure of* I *defines a local inner product on deformation directions — the effective quantum amplitude structure. Second, the basin competition is resolved by the relative heights of* I *at the basin representatives; the ratio formula (19.1) is the unique probability assignment consistent with the additivity of* I *over independent subsystems (a consequence of Requirement 24.2's nonlocal structure) and with normalization. The specific identification* $I[\Phi_k] \propto |\langle\psi_k|\psi\rangle|^2$ *holds in the limit where the effective wave description is valid — the same regime where quantum mechanics applies as an effective theory.* ∎

Remark 19.1.

This derivation does not reduce Phase Theory to quantum mechanics. It shows that quantum mechanics is the effective description in the basin-competition regime, just as classical field theory is the effective description in the smooth regime. The two effective descriptions are compatible — they overlap in the semiclassical regime where both apply — and both are approximations to the underlying Phase Theory dynamics governed by

$I[\Phi]$

. This is the precise sense in which Phase Theory unifies classical and quantum descriptions: not by eliminating one in favor of the other, but by deriving both as regime-appropriate approximations of a single functional structure.

§20. Recovering Emergent Gravity

Proposition 24.7 (Gravity Emergence).

In the large-scale coherent regime — where phase configurations are globally coherent over scales much larger than the coherence length, and the smooth regime conditions of Definition 24.13 are satisfied — the large-scale gradient structure of stable admissible configurations induces an effective Riemannian geometry $(M_{eff}, g_{\mu\nu})$ on the emergent spacetime, satisfying the Einstein field equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_{eff}\, T_{\mu\nu} + R_{\mu\nu}^{(Phase)} \quad (20.1)$$

as a leading-order approximation, where $R_{\mu\nu}^{(Phase)}$ represents sub-leading Phase Theory refractive corrections of the type predicted in Paper 1.

Proof Sketch. *In the large-scale coherent regime, the phase configuration Φ determines a preferred foliation of the emergent space by coherence surfaces. The coherence penalty $C[\Phi]$ restricted to slowly varying configurations takes the form $\int g^{\mu\nu}(\Phi)\, \partial_\mu\varphi^a\, \partial_\nu\varphi^a\, d^4x$, where $g_{\mu\nu}(\Phi)$ is the effective metric induced by the phase configuration. The condition $\delta I = 0$ applied to the topological term under this metric identifies the Einstein tensor of $g_{\mu\nu}$ with the gradient structure of the topological invariants — reproducing the Einstein field equations with an effective gravitational constant G_{eff} determined by the parameters α, β, γ. Sub-leading corrections from the nonlocal components of I produce the refractive gravity deviations of Paper 1.* ∎

The refractive gravity corrections $R_{\mu\nu}^{(Phase)}$ predicted in Paper 1 now have a precise functional-analytic origin: they are the subleading terms in the large-scale expansion of $\delta I = 0$, arising from the nonlocal components of $C[\Phi]$ and

$D[\Phi]$ that are not captured by the local metric approximation. These corrections become significant at very high curvatures (near black hole singularities) and at very large scales (cosmological corrections to general relativity). Both regimes are observationally interesting and provide specific empirical targets for Phase Theory.

> **Remark 20.1 (Background Independence).**
>
> The emergent metric
>
> $g_{\mu\nu}$
>
> is not a background structure — it is derived from the phase configuration
>
> Φ
>
> itself. Different admissible configurations generate different emergent geometries. This means Phase Theory is fundamentally background-independent, consistent with the requirements of general covariance. The "vacuum" geometry (flat space, or de Sitter space with an appropriate cosmological constant) corresponds to the vacuum-like admissible configuration
>
> Φ_{vac}
>
> of Proposition 24.5.

§21. Functional-Analytic Comparison with Standard Approaches

We now place the admissibility functional class in the context of the standard approaches to quantum gravity and unified theories. The comparison is organized by the functional-analytic properties of each approach's central mathematical object.

Property	QFT Action	Regge/ Spinfoam	Polyakov Action	Causal Set	I[Φ] (Phase Theory)
Locality	Local density	Edge-local	Worldsheet-local	Nearest-neighbor	Nonlocal (global)
Topology sensitivity	Weak	Strong	Moderate	Discrete	Strong (required)
Background independence	No	Yes	No	Yes	Yes
Coercivity analog	Partial (unitarity)	Partial	Partial (modular)	None explicit	Explicit (Req. 24.3)
Nonlocality mechanism	None (local)	Global summing	T-duality	None	Global constraint term
Semicontinuity	Assumed smooth	Piecewise	Smooth	Discrete	Explicit (Req. 24.1)

The comparison reveals a clear pattern. Standard QFT action functionals are local, perturbative, and background-dependent — they are the most restricted class. Loop quantum gravity (Regge/spinfoam) achieves background independence and discrete topology sensitivity but lacks a coercivity mechanism and a principled nonlocality structure. String theory (Polyakov action) is background-dependent and achieves nonlocality only through T-duality (a perturbative symmetry), not through global constraints. Causal set theory has the correct discrete, background-independent character but lacks a functional-analytic framework with the properties studied here.

Phase Theory's admissibility functional class is the most general in this comparison, designed specifically to subsume the effective descriptions of all the others in appropriate limits. The QFT action emerges as the local approximation to I in the smooth regime (Section 18). The Regge-like discrete geometry emerges from the topological term $T[\Phi]$ restricted to discrete substrates. The string worldsheet action emerges in a subsector of Phase Theory where the phase degrees of freedom live on extended two-dimensional objects. Causal set combinatorics emerge from the discrete information-density structure of $D[\Phi]$.

Remark 21.1 (Uniqueness of the Phase Theory Setting).

The combination of background independence, explicit coercivity-like bounds, principled nonlocality, and topology sensitivity is unique to the admissibility functional class among the approaches considered. No standard approach achieves all four properties simultaneously. This combination is not a luxury — each property is required by at least two of the six requirements derived from prior papers, and their conjunction is necessary for Phase Theory's full empirical reduction program.

§22. Functional Topology of Padm

Having established the existence and key properties of P_{adm}, we now study its global topological structure as a subspace of $\tilde{P}$.

By Proposition 24.0.1, P_{adm} is a closed subspace of $\tilde{P}$. Its connected components — the path-connected components of P_{adm} — are the admissibility basins B_σ for each topological sector σ. Each basin is itself closed (by Requirement 24.4 and the closed-ness of individual topological sectors in the Phase-natural topology).

Definition 24.15 (Admissibility Boundary).

The *admissibility boundary* of P_{adm} is the set

$$\partial P_{adm} = \{\Phi \in \tilde{P} : I[\Phi] = 0\} \quad (22.1)$$

This is the set of configurations at which the admissibility functional exactly vanishes — marginally admissible configurations.

By the argument of Proposition 24.0.3, ∂P_{adm} is closed. The physical interpretation of ∂P_{adm} is rich and multifaceted:

- **Phase transitions:** Configuration paths that cross ∂P_{adm} correspond to phase transitions in the Phase Theory sense — discontinuous changes in the global consistency of the configuration. These need not correspond to thermodynamic phase transitions, but in appropriate regimes they do.
- **Particle creation and annihilation thresholds:** The creation or annihilation of a particle (topological defect) requires the configuration to pass through a stratum boundary in $\tilde{P}$, which intersects ∂P_{adm}. The admissibility boundary thus encodes the energetic thresholds for particle creation.
- **Topological sector boundaries:** The boundary between distinct topological sectors in $\tilde{P}$ is contained in ∂P_{adm}: crossing a sector boundary takes the configuration through $I[\Phi] = 0$ (since the topological invariants change discretely, and the transition state is marginally admissible).

Remark 22.1 (Non-Manifold Structure).

The boundary

∂P_{adm}

is generically not a smooth manifold. It is the zero-set of a nonlocal, non-smooth functional on a stratified space — which generically produces a stratified set with singularities, corners, and higher-codimension intersections. The study of the geometry of

∂P_{adm}

> — its singularity types, its homology, and its relationship to the stratification of
>
> $\tilde{P}$
>
> — is an important open problem (see Section 26).

§23. Stratified Structure and Singular Configurations

The Phase configuration space $\tilde{P}$ is not a smooth manifold but a stratified space — a union of smooth manifolds (strata) of varying dimensions, glued together along their boundaries in a controlled way. This stratified structure is essential for understanding how particles, defects, and transitions are encoded in the Phase Theory framework.

Definition 24.16 (Stratification of P̃).

A *stratification* of $\tilde{P}$ is a decomposition

$$\tilde{P} = \sqcup_{k\geq 0} \tilde{P}_k \quad (23.1)$$

where $\tilde{P}_0$ is the (dense) smooth stratum of configurations without topological defects, $\tilde{P}_1$ is the stratum of configurations with exactly one topological defect, $\tilde{P}_k$ is the stratum of configurations with exactly k defects, and the closure of $\tilde{P}_k$ contains $\tilde{P}_{k-1}$ (lower strata are limiting cases of higher-defect configurations).

Singular configurations — those lying in lower-dimensional strata $\tilde{P}_k$ for $k \geq 1$ — are precisely the configurations with topological defects: particles, vortices, monopoles, and their Phase Theory analogs. The key statement is that these singular configurations are not separately postulated entities in Phase Theory. They arise naturally as the singularities of the admissibility landscape — the points where the smooth stratum breaks down and the configuration is "most tightly constrained" by the admissibility functional.

Theorem 24.3 (Defects as Singularities).

Every stable topological defect in Phase Theory corresponds to a configuration in the stratum boundary $\tilde{P}_0 \cap closure(\tilde{P}_1)$ — a limiting case of smooth configurations that maintains admissibility in the limit. The admissibility functional I extends continuously (in the lower semicontinuous sense) to all strata, and the defect configurations are local maxima of I restricted to their stratum.

This theorem resolves, at the Phase Theory ontological level, the long-standing conceptual puzzle of particle-field duality. In quantum field theory, particles and fields are described by different mathematical objects — particles as excitations of fields, fields as operator-valued distributions — with the relationship between them given by the non-trivial machinery of second quantization. In Phase Theory, there is a single mathematical object — the phase configuration $\Phi \in \tilde{P}$ — and the particle/field distinction arises from the stratification of $\tilde{P}$: smooth configurations (fields) live in $\tilde{P}_0$, and particle-like configurations (defects) live in $\tilde{P}_1$ and higher strata.

Remark 23.1 (Stratification and Renormalization).

The need for renormalization in quantum field theory is closely connected to the singular behavior of field configurations at short distances. In the Phase Theory stratified setting, the short-distance singularities are regularized by the coercivity-like bound (Requirement 24.3): configurations of infinite local complexity are automatically inadmissible. This suggests that Phase Theory may provide a natural ultraviolet completion of quantum field theory, without the need for ad hoc cutoffs or renormalization group improvements. The renormalization group flow of Section 24 is the Phase Theory analog of the RG flow in QFT, but arises from the structure of the admissibility functional rather than from renormalization of a Lagrangian.

§24. Renormalization-Group-Like Flows on the Functional Space

Having established the properties of individual admissibility functionals, we now consider the space of all admissibility functionals satisfying Requirements 24.1–24.6 and the flow on this space induced by coarse-graining.

Definition 24.17 (Coarse-Graining Flow).

For a scale parameter $\lambda > 0$, the *coarse-graining map* $CG_\lambda : \tilde{P} \to \tilde{P}_\lambda$ identifies all configurations that are indistinguishable at resolution λ (i.e., at scales larger than λ). The *coarse-grained admissibility functional* $I_\lambda : \tilde{P}_\lambda \to \mathbb{R}$ is defined by

$$I_\lambda[\Phi_\lambda] = \sup\{I[\Phi] : CG_\lambda(\Phi) = \Phi_\lambda\} \quad (24.1)$$

The *RG-like flow* is the one-parameter family $\{I_\lambda\}_{\lambda>0}$.

Theorem 24.4 (Closure of the Admissibility Class Under Coarse-Graining).

If I satisfies Requirements 24.1–24.6, then the coarse-grained functional I_λ also satisfies Requirements 24.1–24.6 (with the coarse-grained complexity measure K_λ and coarse-grained topology on $\tilde{P}_\lambda$).

Proof Sketch. *Lower semicontinuity of I_λ: The sup over a closed set of lower semicontinuous functions is lower semicontinuous. Topological compatibility: coarse-graining preserves topological sectors (since topological invariants are large-scale properties and thus survive coarse-graining). Coercivity: the coarse-grained complexity $K_\lambda[\Phi_\lambda] \le K[\Phi]$ for any Φ mapping to Φ_λ, so if $K_\lambda \to \infty$, then $K[\Phi] \to \infty$ for the maximizing Φ, driving $I \to -\infty$ by the original coercivity, hence $I_\lambda \to -\infty$. The remaining requirements follow by analogous arguments.* ∎

The fixed points of the RG-like flow — functionals I_* satisfying $I_{\lambda,*} = I_*$ for all λ — are the scale-invariant admissibility functionals. These correspond to conformal effective theories in the emergent field theory description. The existence and classification of fixed points is a major open problem (Section 26).

Near a fixed point, the linearized flow generates beta-function-like equations governing the running of the effective parameters $\alpha(\lambda)$, $\beta(\lambda)$, $\gamma(\lambda)$ of the representative family (11.1):

$$\lambda\, d\alpha/d\lambda = \beta_\alpha(\alpha, \beta, \gamma),\ \lambda\, d\beta/d\lambda = \beta_\beta(\alpha, \beta, \gamma),\ \lambda\, d\gamma/d\lambda = \beta_\gamma(\alpha, \beta, \gamma) \quad (24.2)$$

where β_α, β_β, β_γ are the Phase Theory beta functions. These govern the scale dependence of the emergent effective coupling constants and reproduce, in the smooth regime, the beta functions of the standard model gauge couplings. The explicit computation of the Phase Theory beta functions from the admissibility functional structure is a priority for Paper 25.

§25. Falsifiability Contributions from the Functional Analysis

The functional-analytic constraints developed in this paper directly strengthen the falsifiability program initiated in Paper 18. The six requirements (24.1–24.6) are not merely mathematical niceties — they generate specific, testable predictions about the physical world. We list five concrete falsification criteria derived from the functional analysis:

(1) Stable defect spectra. Requirement 24.5 (stability) and the Hessian spectral analysis (Section 8) predict that the spectrum of stable topological defects in Phase Theory corresponds to the negative eigenvalue spectrum of the Hessian of I at stable configurations. This spectrum generates a specific particle mass spectrum. If no admissibility functional in the constrained class reproduces the observed particle spectrum of the Standard Model (up to and including corrections from Phase Theory's sub-leading terms), Phase Theory is falsified at this level. The predicted additional particle species (approximately 25 ± 10 species beyond the Standard Model, from Paper 23) must be recoverable from the Hessian spectrum of some I in the class.

(2) Interference statistics. Proposition 24.6 (Born-rule emergence) predicts that interference experiments must yield statistics consistent with the basin overlap measure (19.1). If the basin competition regime generates probability distributions inconsistent with quantum mechanical interference patterns — for example, if it predicts systematically different double-slit fringe patterns — Phase Theory is falsified. This is a stringent test, given the extraordinary precision of quantum optical interference measurements.

(3) Emergent relativistic invariances. Proposition 24.7 (gravity emergence) and Section 18 (classical emergence) jointly predict that the effective dynamics in the smooth regime are Lorentz-invariant and diffeomorphism-invariant to leading order. If the effective equations (18.2) derived from some member of the admissibility functional class violate Lorentz invariance at currently accessible energy scales, that functional — and, if universal, the entire class — is falsified. Current precision tests of Lorentz invariance (from GRB photon arrival times, gamma-ray astronomy, etc.) constrain Lorentz violation to below one part in 10^{20} at the Planck scale, a constraint that the admissibility functional class must satisfy.

(4) Black hole thermodynamic scaling. Theorem 24.2 (saturation) predicts that the admissibility functional class yields the Bekenstein-Hawking entropy formula for black holes as a consequence of the Phase saturation bound. If quantitative comparison with black hole thermodynamics (Hawking radiation, black hole entropy calculations in string theory and loop quantum gravity) reveals inconsistency — specifically, if the Phase Theory prediction for the numerical coefficient in $S_{BH} = A/(4\ell_P^2)$ is not $1/4$ (in Planck units) — the class is falsified. This is a particularly sharp test because the coefficient $1/4$ is known from semiclassical calculations to high precision.

(5) New-particle spectrum prediction. The topological term $T[\Phi]$ (Section 12) and its contribution to the Hessian spectrum (Section 8) predict a specific spectrum of new particles — approximately 25 ± 10 species — with quantum numbers determined by the topological invariants of the Phase substrate. If the admissibility functional class does not generate any such spectrum (i.e., if no choice of parameters α, β, γ and functional form yields a Hessian with the predicted number and quantum-number pattern of stable zero modes), Phase Theory is falsified. Conversely, if future accelerator experiments discover new particles with quantum numbers inconsistent with any topological invariant of the Phase substrate, the theory is also falsified.

Remark 25.1 (Single Failure Suffices).

Any single failure from the five criteria listed above constitutes falsification of Phase Theory at the functional-analytic level — not merely falsification of a particular choice of

I

, but falsification of the claim that the constrained class is empirically adequate. This is a genuine strengthening of the falsifiability status of Phase Theory relative to prior papers, which primarily identified qualitative predictions. The present paper's

> functional-analytic constraints generate quantitative predictions susceptible to precise experimental test.

§26. Open Problems and Research Frontiers

The present paper establishes the structural framework for the admissibility functional. It does not, however, resolve all questions about $I[\Phi]$. We list eight open problems whose resolution would represent significant progress for Phase Theory:

Problem 26.1 (Microscopic Construction). *Explicitly construct the admissibility functional* $I[\Phi]$ *from first principles of the Phase substrate defined in Paper 1.* This is the most fundamental open problem. The present paper establishes the requirements that any such construction must satisfy; the construction itself — deriving I from the substrate ontology rather than postulating its general form — remains to be accomplished. Progress here would mean that Phase Theory achieves the same status as general relativity with respect to its founding equations: derived from a clear geometric principle, not merely postulated. Paper 25 will initiate this construction program.

Problem 26.2 (Uniqueness Up to Parameters). *Is the admissibility functional class defined by Requirements 24.1–24.6 sufficient to determine* I *up to the three parameters* α, β, γ *of the representative family?* Specifically: does every functional satisfying all six requirements take the form (11.1) up to higher-order corrections suppressed by powers of the Phase saturation scale? Uniqueness up to parameters would be an enormously powerful result — it would mean that Phase Theory has only three free parameters (analogous to the three fundamental constants $\hbar, c, G$ of standard physics) and that all other physical quantities are derivable from them. Progress here would require developing a classification theory for functionals satisfying Requirements 24.1–24.6, analogous to the Lovelock theorem for gravitational Lagrangians.

Problem 26.3 (Fréchet Differentiability). *Can the admissibility functional $\mathcal{I}$ be shown to be Fréchet-differentiable on a dense subspace of $\tilde{P}$?* Fréchet differentiability is the natural analog of smooth differentiability for functionals on infinite-dimensional spaces. It would enable a full development of the calculus of variations for $\mathcal{I}$, including rigorous derivation of the Euler-Lagrange equations and control of higher-order corrections. The main obstacle is the stratified, non-linear structure of $\tilde{P}$. Progress here would require new techniques from nonsmooth analysis and geometric measure theory, adapted to the Phase-Theory setting.

Problem 26.4 (Spectral Completeness of the Hessian). *Is the spectrum of the Hessian operator $H[\Phi_0]$ at stable admissible configurations always discrete (i.e., purely point spectrum with no continuous part)?* Discreteness of the spectrum is required for the interpretation of eigenvalues as particle masses — a continuous spectrum would correspond to a continuum of particle species, inconsistent with observation. Progress here would require proving a spectral gap theorem for $H[\Phi_0]$, analogous to the spectral gap results for Schrödinger operators in quantum mechanics. The key tool would likely be a coercivity estimate for the Hessian, derived from Requirement 24.3.

Problem 26.5 (Classification of Topological Sectors). *Given a specific admissibility functional $\mathcal{I}$ satisfying Requirements 24.1–24.6, classify all topological sectors admitted by $\mathcal{I}$ — that is, determine the homotopy type of $\tilde{P}^{sm}$ and the set $\pi_0(\tilde{P}^{sm})$ of connected components.* This is the mathematical foundation of the particle classification program of Paper 3. Progress requires computing the homotopy groups of the Phase configuration space — a problem that depends on the specific choice of Phase substrate and value space, and that may require new techniques in infinite-dimensional homotopy theory.

Problem 26.6 (Fixed-Point Structure of the RG Flow). *How many fixed points does the coarse-graining flow of Section 24 have, and what is their basin of attraction?* Fixed points correspond to scale-invariant (conformal) effective theories. In standard QFT, the classification of conformal fixed points is equivalent to the classification of conformal field theories — a massive program that has occupied mathematical physicists for decades. The Phase Theory analog may be more tractable due to the explicit coercivity structure. Progress here would yield predictions for the universality classes of Phase Theory effective theories and their connection to the observed conformal invariances in statistical mechanics and critical phenomena.

Problem 26.7 (Standard Model Gauge Group from T[Φ]). *Show that the topological term $T[\Phi]$, when restricted to appropriate topological sectors of the Phase configuration space, generates gauge symmetries equivalent or isomorphic to the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$.* This is the deepest connection between Phase Theory and known particle physics. Paper 3 began the program of identifying topological invariants with quantum numbers; the present paper establishes that $T[\Phi]$ is the functional origin of these quantum numbers. The remaining step is to show that the *symmetries* of $T[\Phi]$ in the appropriate sector reproduce the Standard Model gauge symmetries. Progress here would be one of the most significant results in Phase Theory — a derivation of the Standard Model from first principles.

Problem 26.8 (Numerical Approximation). *Develop algorithms for numerically approximating $I[\Phi]$ to sufficient precision for direct comparison with experimental results.* The functional I is defined on an infinite-dimensional space with nonlocal, topological structure — its direct numerical approximation is highly nontrivial. Progress here would require developing finite-dimensional discretizations of $\tilde{P}$ that preserve topological structure (analogous to lattice field theory discretizations that preserve gauge invariance), combined with efficient optimization algorithms for finding stable admissible configurations and computing Hessian spectra. This problem connects Phase Theory to computational physics and mathematical software development, and its resolution would enable quantitative comparison between Phase Theory predictions and experimental data.

§27. Conclusion

This paper has undertaken a systematic functional-analytic treatment of the admissibility functional $I[\Phi]$, the central mathematical object of Phase Theory. Beginning from the accumulated demands of prior papers in the series and proceeding through a careful derivation of necessary structural conditions, we have established that $I[\Phi]$ belongs to a precisely constrained class of functionals — not arbitrary, not uniquely determined, but tightly specified by six requirements whose conjunction is both necessary and sufficient for Phase Theory's full program of empirical reduction.

The six requirements — lower semicontinuity (24.1), local sensitivity with global dependence (24.2), coercivity-like bounds (24.3), topological compatibility (24.4), stability under admissible perturbations (24.5), and constraint nonlocality (24.6) — are individually motivated by specific prior results and collectively form a coherent and internally consistent set of conditions. We have proved that these requirements are compatible: the admissible functional class is non-empty, demonstrated by explicit construction of the representative family $I[\Phi] = \alpha T[\Phi] - \beta C[\Phi] - \gamma D[\Phi]$, verified against all six requirements.

From this framework, we have derived a cascade of consequences. The closedness of the admissible space (Proposition 24.0.1), the non-emptiness of admissible basins (Propositions 24.1 and 24.5), the stability of topological invariants (Proposition 24.0.3), the no-signaling theorem (Theorem 24.1), and the reduction of Phase saturation bounds to theorems (Theorem 24.2) are all consequences of the six requirements rather than independent postulates.

This represents a net reduction in the axiomatic complexity of Phase Theory, strengthening its theoretical foundation.

The emergent effective theories — classical field theory (Section 18), quantum-mechanical statistics (Section 19), and general-relativistic gravity (Section 20) — all arise as regime-specific approximations to the dynamics generated by the extremal condition $\delta I[\Phi] = 0$. The three approximations are compatible and overlap in appropriate semiclassical limits, providing a coherent picture of the spectrum of known physics emerging from a single functional structure. The corrections to each effective theory — sub-leading admissibility terms — are specific predictions of Phase Theory, distinguishable from the predictions of the Standard Model and general relativity at energy scales approaching the Phase saturation scale.

The comparison with standard approaches (Section 21) reveals that Phase Theory's admissibility functional class is the most general functional framework considered — subsuming QFT actions, Regge/spinfoam sums, string worldsheet actions, and causal set actions as special cases or limits. The combination of background independence, explicit coercivity, principled nonlocality, and full topology sensitivity is unique to the Phase Theory setting and is required by the full empirical reduction program.

The falsifiability program (Section 25) connects the functional-analytic results directly to experiment. Five specific falsification criteria — covering particle spectra, interference statistics, Lorentz invariance, black hole thermodynamics, and new-particle predictions — provide concrete empirical targets for Phase Theory. These are not qualitative predictions but quantitative ones: the Hessian spectrum must yield approximately 25 ± 10 new particle species; the thermodynamic scaling must reproduce the Bekenstein-Hawking coefficient of $1/4$; the Lorentz violation must be below current experimental bounds. Any single failure suffices to falsify the theory at the functional-analytic level.

Eight open problems (Section 26) chart the research frontier: from the explicit microscopic construction of *I[Φ]*, through its uniqueness and differentiability properties, to the classification of topological sectors, the RG fixed-point structure, the derivation of the Standard Model gauge group, and the development of numerical approximation algorithms. These problems span pure mathematics, mathematical physics, and computational physics, and their pursuit will require collaboration across these disciplines.

Papers 25 and beyond will pursue several of these open problems in sequence. Paper 25 will initiate the explicit construction program for *I[Φ]* from microscopic substrate principles. Paper 26 will address the uniqueness question. Paper 27 is expected to derive the Standard Model gauge group from the topological sector classification — a result that, if achieved, will represent Phase Theory's most definitive contact with known physics.

The trajectory of this series — from the initial ontological postulates of Paper 1, through the mathematical development of the phase configuration space in Papers 2–23, to the functional-analytic treatment of the present paper — traces the maturation of a unified physical theory from speculation to mathematical precision. What began as an intuition about phase and consistency has become a framework with theorems, predictions, and falsification criteria of real scientific substance.

Admissibility is not a metaphor. It is a functional class with teeth.

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